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MATHEMATICS

STANDARD SIX

Terminal exam - May 2024
Answer Key and Worked Solutions

Subject	Mathematics
Code	Not printed in source
Level	Standard Six
Year	2024
Exam Board / Issuer	MESP Tanzania
Exam	Terminal exam - May
Type	Answer key and worked solutions
Questions	34 numbered questions/subparts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S6-2024-MESP-Tanzania-May-Terminal-exam.pdf

Notes about the source paper

Items with missing information or printing errors are explained in their solutions. A model response is an example, rather than a prescribed official answer.

Calculations

1(i). Add 90000 + 68000.

Step 1: 1 place: $0 + 0 = 0$; write 0.

Step 2: 10 place: $0 + 0 = 0$; write 0.

Step 3: 100 place: $0 + 0 = 0$; write 0.

Step 4: 1000 place: $0 + 8 = 8$; write 8.

Step 5: 10000 place: $9 + 6 = 15$; write 5, carry 1.

Step 6: Write final carry 1 in the next place.

Step 7: $90000 + 68000 = 158000$.

Final Answer: 158000

1(ii). Subtract 100000 - 1001.

Step 1: 1 place: regroup 1 from the next place; $10 - 1 = 9$.

Step 2: 10 place: regroup 1 from the next place; $9 - 0 = 9$.

Step 3: 100 place: regroup 1 from the next place; $9 - 0 = 9$.

Step 4: 1000 place: regroup 1 from the next place; $9 - 1 = 8$.

Step 5: 10000 place: regroup 1 from the next place; $9 - 0 = 9$.

Step 6: 100000 place: $0 - 0 = 0$.

Step 7: $100000 - 1001 = 98999$; check $98999 + 1001 = 100000$.

Final Answer: 98999

1(iii). Multiply 6800 x 25.

Step 1: Split into place values: $6000 \times 25 = 150000$; $800 \times 25 = 20000$.

Step 2: Add the partial products: 170000.

Final Answer: 170000

1(iv). Divide 775 by 15.

Step 1: $77 \div 15$: write 5; $5 \times 15 = 75$; $77 - 75 = 2$.

Step 2: $25 \div 15$: write 1; $1 \times 15 = 15$; $25 - 15 = 10$.

Step 3: $15 \times 51 = 765$.

Step 4: $775 \div 15 = 155/3$; whole-number division: 51 remainder 10.

Final Answer: 155/3

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Mathematics

1(ix). Multiply $(-280) \times (-15)$.

Step 1: The product of two negative numbers is positive.

Step 2: $280 \times 15 = 4200$.

Final Answer: 4200

1(v). Add $9 \frac{1}{2} + 6 \frac{3}{4}$.

Step 1: $9 \frac{1}{2} = \frac{38}{4}$; $6 \frac{3}{4} = \frac{27}{4}$.

Step 2: $38 + 27 = 65$; $65/4 = 16 \frac{1}{4}$.

Final Answer: 16 1/4

1(vi). Subtract $12 \frac{3}{7} - 6 \frac{2}{12}$.

Step 1: $6 \frac{2}{12} = 6 \frac{1}{6}$.

Step 2: Use the common denominator 42: $522/42 - 259/42$.

Step 3: $522 - 259 = 263$; $263/42 = 6 \frac{11}{42}$.

Final Answer: 6 11/42

1(vii)-divide. Divide $\frac{8}{9}$ by $\frac{2}{4}$.

Step 1: Multiply $\frac{8}{9}$ by $\frac{4}{2}$.

Step 2: $\frac{32}{18} = \frac{16}{9} = 1 \frac{7}{9}$.

Final Answer: $1 \frac{7}{9}$

1(vii)-multiply. Multiply $7 \frac{3}{5} \times 4 \frac{2}{3}$.

Step 1: Convert the mixed numbers to $\frac{38}{5}$ and $\frac{14}{3}$.

Step 2: $38 \times 14 = 532$; $5 \times 3 = 15$.

Step 3: $\frac{532}{15} = 35 \frac{7}{15}$.

Final Answer: $35 \frac{7}{15}$

1(viii). Multiply 10.26×3.4 .

Step 1: Remove the decimal points: $1026 \times 34 = 34884$.

Step 2: Restore 3 decimal places.

Final Answer: 34.884

1(x). Add $9.0855 + 4.2040$.

Step 1: Align the 4 decimal places.

Step 2: Ten-thousandths: $5 + 0 = 5$; thousandths: $5 + 4 = 9$; hundredths: $8 + 0 = 8$; tenths: $0 + 2 = 2$; whole-number part: $9 + 4 = 13$.

Final Answer: 13.2895

2(i). Convert $\frac{2}{5}$ to a percentage.

Step 1: $\frac{2}{5} = 0.4$.

Step 2: $0.4 \times 100 = 40$.

Final Answer: 40%

2(ii). Find the square of 49.

Step 1: $49 \times 49 = 2401$.

Final Answer: 2401

2(iii). Find the LCM of 9, 15 and 18.

Step 1: $9 = 3^2$; $15 = 3 \times 5$; $18 = 2 \times 3^2$.

Step 2: Use the highest prime powers: $2 \times 3^2 \times 5 = 90$.

Final Answer: 90

2(iv). Find the GCF of 59, 48 and 18.

Step 1: 59 is prime and divides neither 48 nor 18.

Step 2: The only common factor is 1.

Final Answer: 1

2(v). Convert 145% to a fraction.

Step 1: Write $145/100$ and divide the numerator and denominator by 5.

Step 2: $145/100 = 29/20 = 1 \frac{9}{20}$.

Final Answer: 29/20

2(vi). List all odd numbers between 54 and 56.

Step 1: 55 is the only integer strictly between these numbers.

Final Answer: 55

3(i). Find the sum of the first 6 positive even numbers.

Step 1: $2 + 4 + 6 + 8 + 10 + 12 = 42$.

Step 2: If the sequence starts at 0 instead, the sum is $0 + 2 + 4 + 6 + 8 + 10 = 30$.

Final Answer: 42 for positive even numbers (30 if including 0).

3(ii). Subtract DV - CD and give the answer in Arabic numerals.

Step 1: $DV = 505$; $CD = 400$.

Step 2: $505 - 400 = 105$.

Final Answer: 105

3(iii). Find the difference between the greatest and least prime numbers between 10 and 20.

Step 1: The prime numbers are 11, 13, 17 and 19.

Step 2: $19 - 11 = 8$.

Final Answer: 8

4(i). Find the mean of 15, 35, 0, 40 and 65.

Step 1: $15 + 35 + 0 + 40 + 65 = 155$.

Step 2: $155 / 5 = 31$.

Final Answer: 31

4(ii). How many lines of symmetry does a non-square rectangle have?

Step 1: Its lines of symmetry are the horizontal and vertical midlines.

Final Answer: 2

4(iii). Find the difference between the areas of a 15 cm by 30 cm rectangle and a triangle with height 15 cm and base 30 cm.

Step 1: Rectangle area: $15 \times 30 = 450 \text{ cm}^2$.

Step 2: Triangle area: $15 \times 30 / 2 = 225 \text{ cm}^2$.

Step 3: Difference: $450 - 225 = 225$.

Final Answer: 225 cm²

5(i). Add 300 shillings to 12% of 3600 shillings.

Step 1: $3600 \times 12 / 100 = 432$.

Step 2: $432 + 300 = 732$.

Final Answer: Sh 732

5(ii). Subtract 3 hours 55 minutes from 4 hours 13 minutes.

Step 1: Borrow an hour to write the larger duration as 3 hours 73 minutes.

Step 2: $73 - 55 = 18$ minutes; hours: $3 - 3 = 0$.

Final Answer: 18 minutes

5(iii). What is the hundred-thousands digit in 4761592?

Step 1: The digits are 4 in millions, 7 in hundred-thousands and 6 in ten-thousands.

Final Answer: 7

6(i). Find y from the printed chain $100 = 2 \times y = 10$.

Step 1: A chained equality requires $100 = 10$, which is false.

Step 2: If $100 = 2y$ is intended, $y = 50$; if the part $2y = 10$ is intended, $y = 5$.

Final Answer: The printed chain has no solution. Conditional answers are $y = 50$ or $y = 5$, depending on the correction.

6(ii). Two bells ring every 7 and 11 minutes. When will they next ring together?

Step 1: Both numbers are prime.

Step 2: $\text{LCM} = 7 \times 11 = 77$.

Final Answer: 77 minutes

6(iii). How many $\frac{1}{4}$ parts make $\frac{1}{2}$?

Step 1: $(\frac{1}{2}) \div (\frac{1}{4}) = 2$.

Final Answer: 2

7(i). A farmer harvests 25 sacks of beans, 65 of maize, 55 of rice and 35 of groundnuts, then sells $\frac{1}{5}$ of each crop. How many sacks are sold?

Step 1: $25 + 65 + 55 + 35 = 180$ sacks.

Step 2: $180 \div 5 = 36$.

Final Answer: 36 sacks

7(ii). Start with Sh 10000 and buy 3 cakes at Sh 500 each, 4 fruits at Sh 450 each and 3 kg of sugar at Sh 2200 per kg. Find the balance.

Step 1: $3 \times 500 = 1500$; $4 \times 450 = 1800$; $3 \times 2200 = 6600$.

Step 2: $1500 + 1800 + 6600 = 9900$.

Step 3: $10000 - 9900 = 100$.

Final Answer: Sh 100

8(i). An isosceles triangle has apex angle 100° . Angle b is formed by the extended base and a ray parallel to the left side. Find b .

Step 1: The base angles are equal; $180 - 100 = 80$.

Step 2: $80 / 2 = 40$.

Step 3: The corresponding angle is $b = 40$.

Final Answer: 40°

8(ii). A right triangle has hypotenuse 13 cm and base 5 cm. Find its area.

Step 1: $\text{Height}^2 = 13^2 - 5^2 = 169 - 25 = 144$, so the height is 12 cm.

Step 2: $\text{Area} = 5 \times 12 / 2 = 30$.

Final Answer: 30 cm^2

8(iii). Find the exterior angle f at the left vertex. Only the lower interior angle of 78° is given.

Step 1: Let the upper interior angle be t .

Step 2: The exterior angle f is the sum of the opposite interior angles: $78^\circ + t$.

Step 3: No value for t is printed. The adjacent angle at the lower crossing is 102° , but that is not f .

Final Answer: $f = 78^\circ + t$; it cannot be uniquely determined without the upper angle.

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Source Checks

Darasa Huru - MESP Tanzania Standard Six Terminal Exams May 2024